Higher National Diploma in Information Technology
First Year, Second Semester Examination-2021
HNDIT1211: Data Structures and Algorithms

Instructions for Candidates:
Answer four (04) questions only.

No. of questions : 05
No. of pages : 03
Time : Two hours

Question 01

i. Define the term Data Structure. (02 Marks)

ii. Briefly explain the importance of using data structures in programming. (04 Marks)

iii. Assume that you have designed an algorithm to do a specific task in your final project. Briefly explain how you can evaluate the efficiency of your algorithm. (04 Marks)

iv. Write a C++ function that calculate and returns the sum of all the even numbers of a given array. (07 Marks)

v. Write a C++ program that finds the maximum and minimum elements in the following array.

78	65	12	48	23	96	77	18	29	54

(08 Marks)

Question 02

i. What is an abstract data type? (02 Marks)

ii. State whether the following data structures is leaner or nonlinear. (04 Marks)

a. **Array - Linear**
b. **Stack - Linear**
c. **Queue Linear**
d. **Graph – Non-Linear**

iii. Compare and contrast arrays and linked lists data structures, highlighting their advantages and disadvantages. (06 Marks)

iv. Declare a C++ structure to store an integer and a String. (06 Marks)

v. Implement a C++ function to insert a node at the beginning of a linked list. (07 Marks)

Question 03

i. Differentiate Stack and Queue data structures. (04 Marks)

ii. Graphically represent the static and dynamic implementations of Stack? (06 Marks)

iii. Illustrate the **graphical representation** and **implementation** of following Stack operations. (07 Marks)

a. initializeStack()
b. push(5)
c. y = topElement( )
d. x= pop()

iv. Graphically illustrate the static implementation of the following Queue operations **(Size of the array is 5).** (08 Marks)

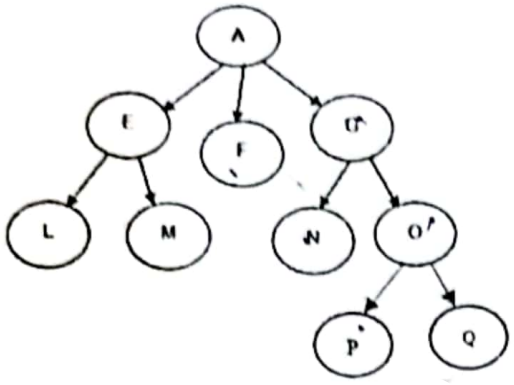
a. initializeQueue()
b. enQueue (25)
c. enQueue (70)
d. deQueue ()
e. q = isFull()
f. x = deQueue ()
g. x = firstElement ()
h. enQueue (25)

Question 04

i. Define Tree data structure. (02 Marks)

ii. Compare Binary Tree and Binary Search Tree with a suitable diagrams. (06 marks)

iii. Give the following details related to the given tree diagram. (09 Marks)



a. Size of the Tree
b. Depth of Node 'N
c. Height of the Tree
d. Degree of a Node 'P'
e. Degree of the Tree
f. Path to node 'M'
g. Leaf Nodes

iv. Diagrammatically show the insertion of following data set to a binary search tree.

a. 9, 30, 24, 10, 37,18,33,5 (08 Marks)

Question 05

i. Define the term 'sorting 'giving at least one example. (03 Marks)

ii. Consider the following data set.

Data Set: 94, 40, 62, 2, 18

a. Sort above data set using selection sort. (05 Marks)

b. Sort the above data set using bubble sort. (05 Marks)

iii. Write down the pseudo codes to implement a linear search algorithm. (05 marks)

iv. Write a C++ function to implement a linear search in an integer array. (07 Marks)